

Control Flow: Conditional Branch

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Control Flow

- Conditional Branch
 - if / switch
- Loop
 - for / while

Outline

- if / else if / else
- switch

Outline

- if / else if / else
- switch

if

- Example 1

```
4  int main() {  
5  
6      int i = 3, j = 5;  
7      if (i < j && j < 7) {  
8          cout << "Hello World\n";  
9      }  
10     if (i < j) {  
11         if (j < 7) {  
12             cout << "Hello World\n";  
13         }  
14     }  
15 }
```

if

- Example 2

```
4  int main() {  
5  
6      int i = 3, j = 5, k = 7;  
7      if (i > j || j <= k || (i+j) < 7) {  
8          cout << "Pass\n";  
9      }  
10 }
```

Practice

- (Practice 1) Rewrite 'Example 2' by using 3 if statements
- (Practice 2) Rewrite 'Example 3' by removing all logical operators

if

- Example 3

```
4  int main() {  
5  
6      int i = 3, j = 5, k = 7;  
7      if ((i > j || j <= k) && (i+j) > 7) {  
8          cout << "Hello\n";  
9      }  
10     return 0;  
11 }
```



```
if ((err = ReadyHash(&SSLHashSHA1, &hashCtx)) != 0)
    goto fail;
if ((err = SSLHashSHA1.update(&hashCtx, &clientRandom)) != 0)
    goto fail;
if ((err = SSLHashSHA1.update(&hashCtx, &serverRandom)) != 0)
    goto fail;
if ((err = SSLHashSHA1.update(&hashCtx, &signedParams)) != 0)
    goto fail;
    goto fail; /* MISTAKE! THIS LINE SHOULD NOT BE HERE */
if ((err = SSLHashSHA1.final(&hashCtx, &hashOut)) != 0)
    goto fail;

err = sslRawVerify(...);

. . .
```

<https://nakedsecurity.sophos.com/2014/02/24/anatomy-of-a-goto-fail-apples-ssl-bug-explained-plus-an-unofficial-patch/>

老師的建議：
只有一行敘述也請加上小括弧

else

```
4  int main() {  
5  
6      int i = 3, j = 5, k = 7;  
7  
8      if ((i+j) < 7) {  
9          cout << "Hello\n";  
10     } else {  
11         cout << "World\n";  
12     }  
13 }
```

else if

```
4  int main() {  
5  
6      int score = 80;  
7  
8      if (score >= 90) {  
9          cout << ">=90\n";  
10     } else if (score >= 80) {  
11         cout << ">=80\n";  
12     } else if (score >= 70) {  
13         cout << ">=70\n";  
14     }  
15 }
```

老師的提醒：
電腦科學是一門「人類定義」的科學

if / else if / else

```
4  int main() {  
5  
6      int score = 70;  
7  
8      if (score >= 80) {  
9          cout << ">=80\n";  
10     } else if (score >= 60) {  
11         cout << ">=60\n";  
12     } else {  
13         cout << "< 60\n";  
14     }  
15 }
```

Practice

- FizzBuzz
 - <http://www.kalzumeus.com/2011/10/28/dont-call-yourself-a-programmer/>
 - Many people already successfully employed as senior engineers cannot actually implement FizzBuss
 - "FizzBuzz Questions" named after a game children often play (or are made to play) in schools in the UK. An example of a Fizz-Buzz question is the following:
 - Write a program that prints the numbers from 1 to 100. But for multiples of three print "Fizz" instead of the number and for the multiples of five print "Buzz". For numbers which are multiples of both three and five print "FizzBuzz".
 - Most good programmers should be able to write out on paper a program which does this in a under a couple of minutes. Want to know something scary? **The majority of comp sci graduates can't. I've also seen self-proclaimed senior programmers take more than 10-15 minutes to write a solution.**

Modern C++ (C++17)

- `if (initialization; condition) ...`

```
4  ∨ int main() {  
5  
6  ∨     if (int i = 3, j = 5, k = 7; (i > j || j <= k) && (i+j) > 7) {  
7      |         cout << "Hello\n";  
8      |     }  
9      |     return 0;  
10 }  
    }
```

```
[lincyu@Chih-YudeMacBook-Pro Chapter4 % clang++ -std=c++17 4_10_ModernCPP_if.cpp  
[lincyu@Chih-YudeMacBook-Pro Chapter4 % ./a.out  
Hello
```

Outline

- if / else if / else
- switch

switch

- Character (Example 1)

```
6      char c = 'B';
7
8      switch (c) {
9          case 'A':
10             cout << "Great\n";
11             break;
12          case 'B':
13             cout << "Good\n";
14             break;
15          case 'C':
16             cout << "OK\n";
17      }
```


switch

- Integer (Example 2)

```
6      int i = 5;
7
8      switch (i) {
9          case 5:
10             cout << "Great\n";
11             break;
12          case 4:
13             cout << "Good\n";
14             break;
15          case 3:
16             cout << "OK\n";
17      }
```

break

- Example 3

```
6      char c = 'B';
7
8      switch (c) {
9          case 'A':
10             cout << "Great\n";
11          case 'B':
12             cout << "Good\n";
13          case 'C':
14             cout << "OK\n";
15      }
```

default

- Example 4

```
6      char c = 'Z';
7
8      switch (c) {
9          case 'A':
10             cout << "Great\n";
11             break;
12          case 'B':
13             cout << "Good\n";
14             break;
15          case 'C':
16             cout << "OK\n";
17             break;
18          default:
19             cout << c << "\n";
20     }
```

Modern C++ (C++17)

- `switch (initialization; condition) {...}`

```
6      switch (int i = 5; i) {
7          case 5:
8              cout << "Great\n";
9              break;
10         case 4:
11             cout << "Good\n";
12             break;
13         case 3:
14             cout << "OK\n";
15     }
16 }
```

Modern C++ (C++17)

- `[[fallthrough]]`
 - <https://en.cppreference.com/w/cpp/language/attributes/fallthrough>

Practice

- (Practice 1) Rewrite 'Example 4' by using if statement